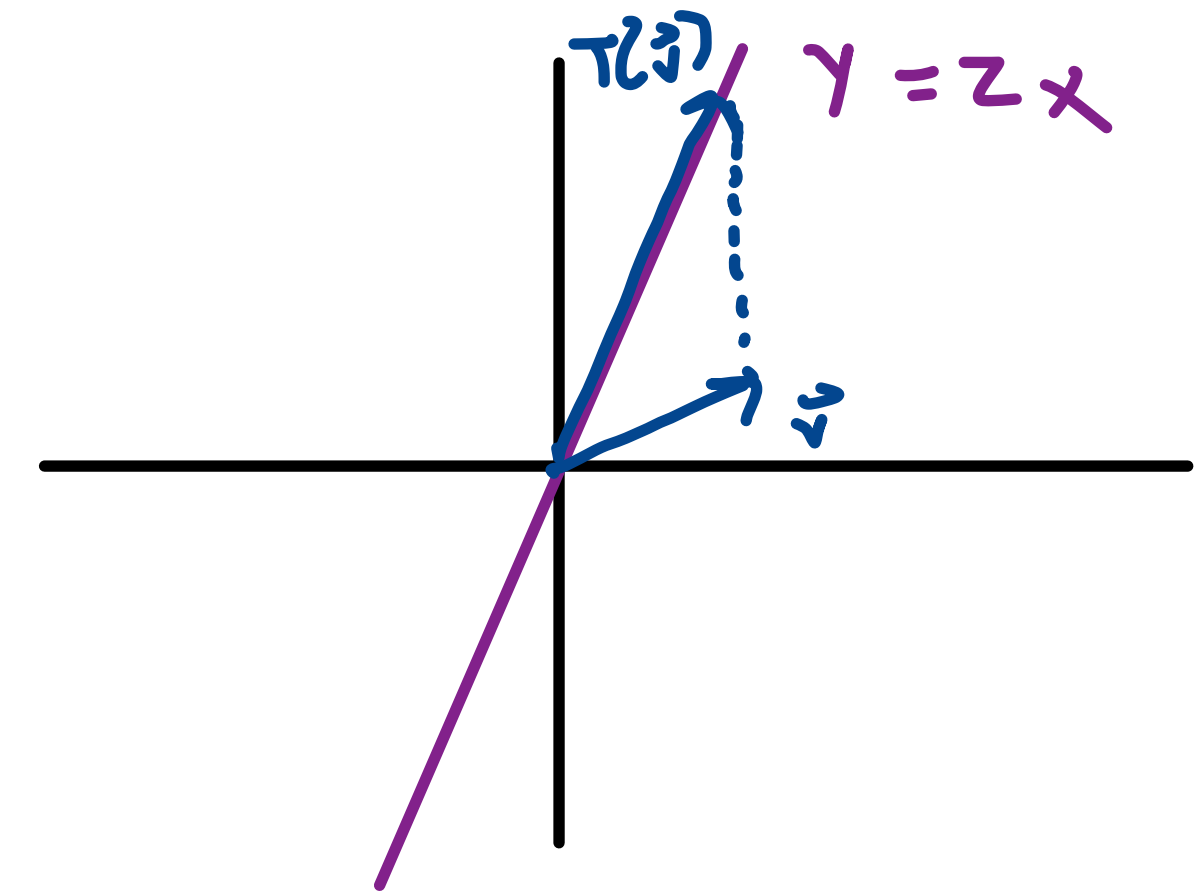


Matrix Algebra

Geometric Algorithms Lecture 9

Practice Problem



Determine the matrix that implements the transformation which projects vectors in $\mathbb{R}^2$ vertically onto the line $y = 2x$ in $\mathbb{R}^2$

(Show that the transformation which projects onto $y = 2x + 3$ is not linear)

Answer

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$$

2×2

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix}$$

$$\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

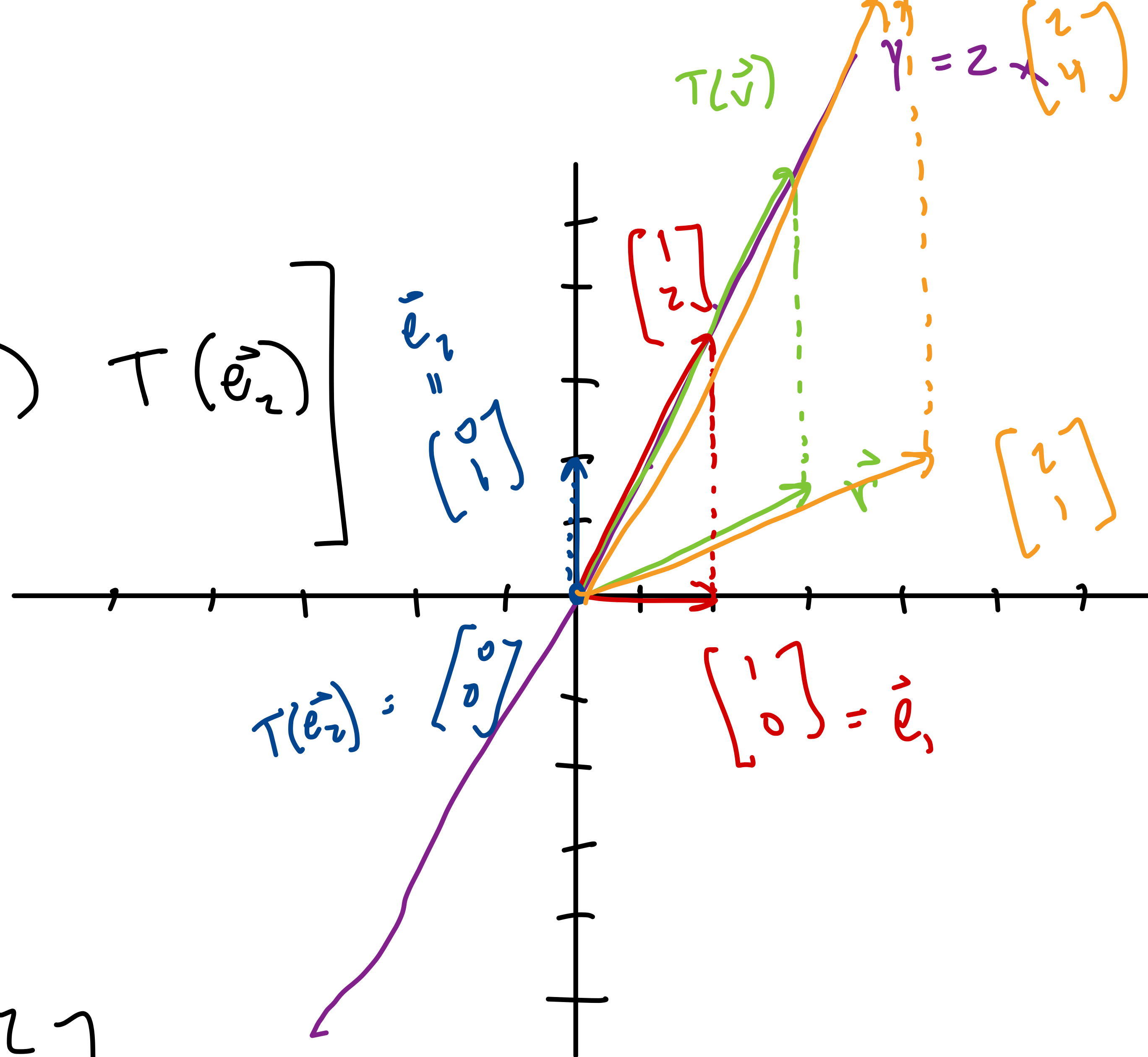
$$T(\vec{e}_2) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$T(\vec{v})$$

$$\gamma = 2 \times \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

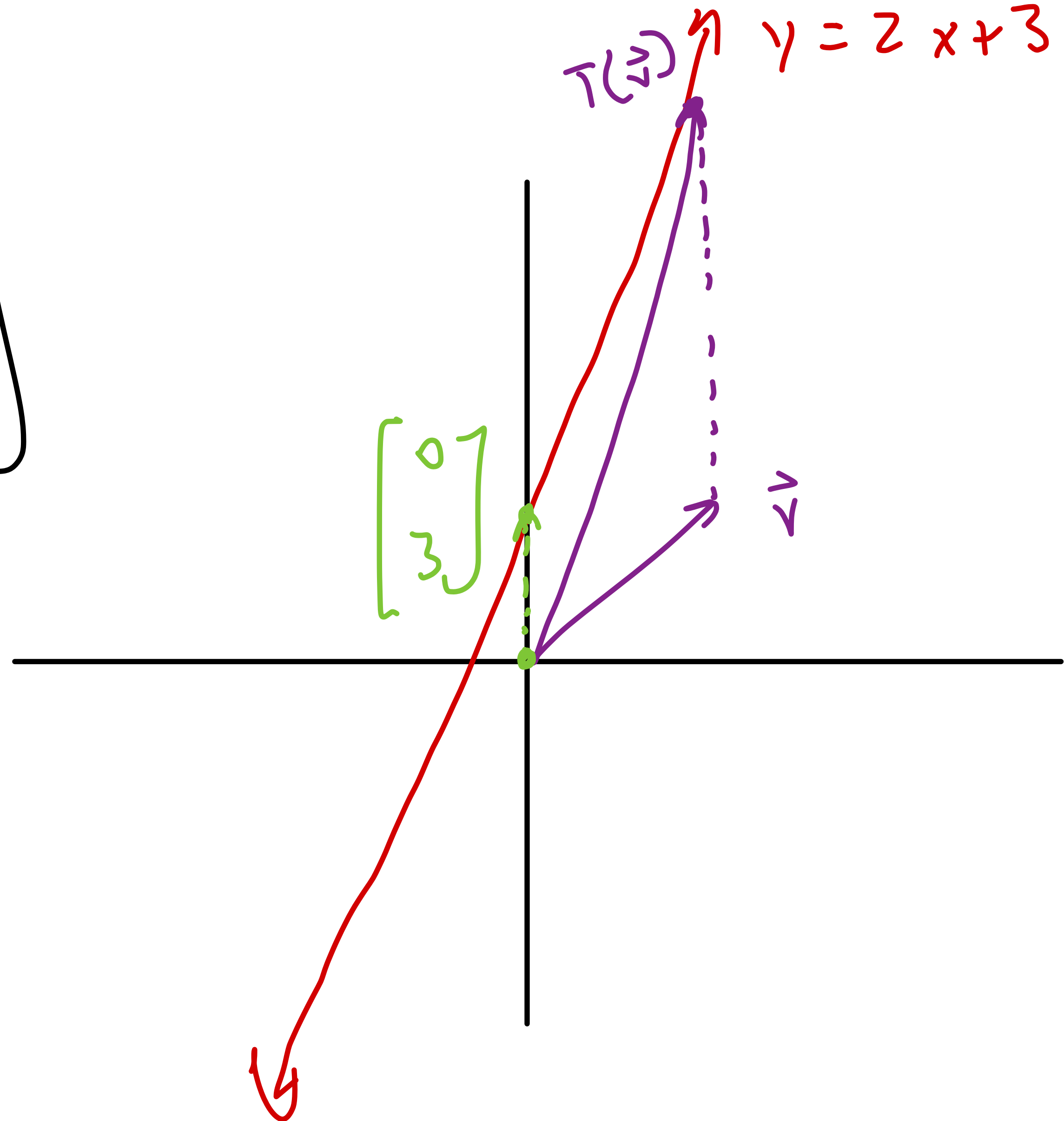
$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \vec{e}_1$$



Answer

$$T\left(\begin{bmatrix} 0 \\ 6 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$



Objectives

- » Connect questions about **matrix equations** and **linear transformations**
- » Motivate **matrix multiplication**
- » Define matrix multiplication
- » Look at the **algebra** of matrix multiplication

Keywords

one-to-one transformation

onto transformation

matrix multiplication

row-column rule

matrix addition and scaling

non-commutativity

Recap

Recap: Matrix Transformations

Matrices *transform* vectors into vectors the span of its columns

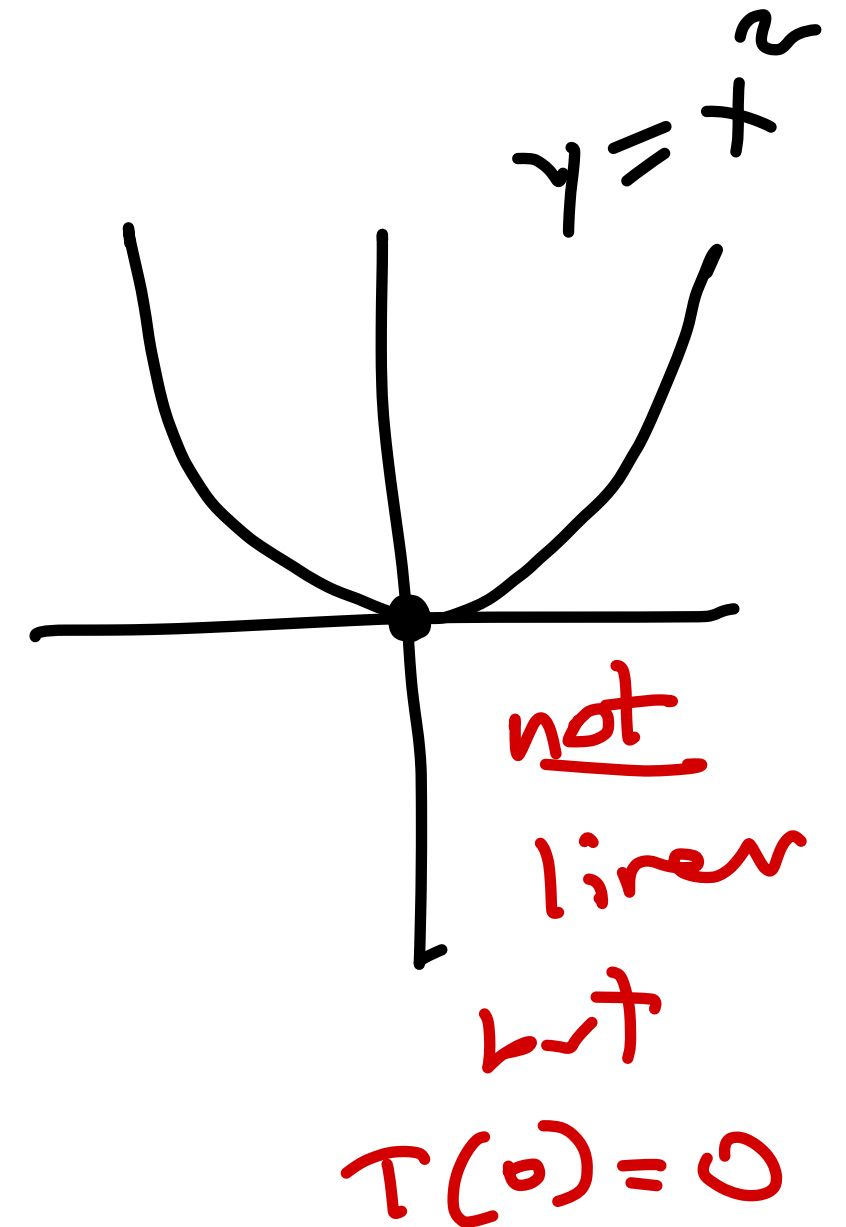
$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

Recap: Linear Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)
2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)



Matrix transformations are linear transformations

Recap: Matrix of Linear Transformation

Thm. A transformation T is linear if and only if it is a matrix transformation (there is a matrix that "implements" T)

Linear transformations are exactly matrix transformations

Recap: Matrix of Linear Transformation

Thm. For any linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$,
the matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)]$$

is the unique matrix that implements T

How do matrix properties relate
to transformation properties?

One-to-One and Onto

Recap: New Interpretations

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

More New Interpretations

Does $A\mathbf{x} = \mathbf{b}$ have a solution for any choice of $\mathbf{b}$?

Full Span

Does $A\mathbf{x} = \mathbf{0}$ have a unique solution?

Lin. Ind.

More New Interpretations

Does $A\mathbf{x} = \mathbf{b}$ have at least one solution for any choice of $\mathbf{b}$?

Does $A\mathbf{x} = \mathbf{b}$ have at most one solution for any choice of $\mathbf{b}$?

Why?

$A\mathbf{x} = \mathbf{0}$ has a
unique solution

($\Rightarrow$)

$$A\vec{v}_1 = \vec{b} \quad A\vec{v}_2 = \vec{b} \\ \vec{v}_1 \neq \vec{v}_2$$

($\Leftarrow$)

$$A\vec{v} = \vec{0} \quad \vec{v} \neq \vec{0}$$

$\equiv$

$A\mathbf{x} = \mathbf{b}$ has at most one
solution

$$A(\underbrace{\vec{v}_1 - \vec{v}_2}_{\text{nontrivial solution}}) = A\vec{v}_1 - A\vec{v}_2 \\ = \vec{b} - \vec{b} \\ = \vec{0}$$

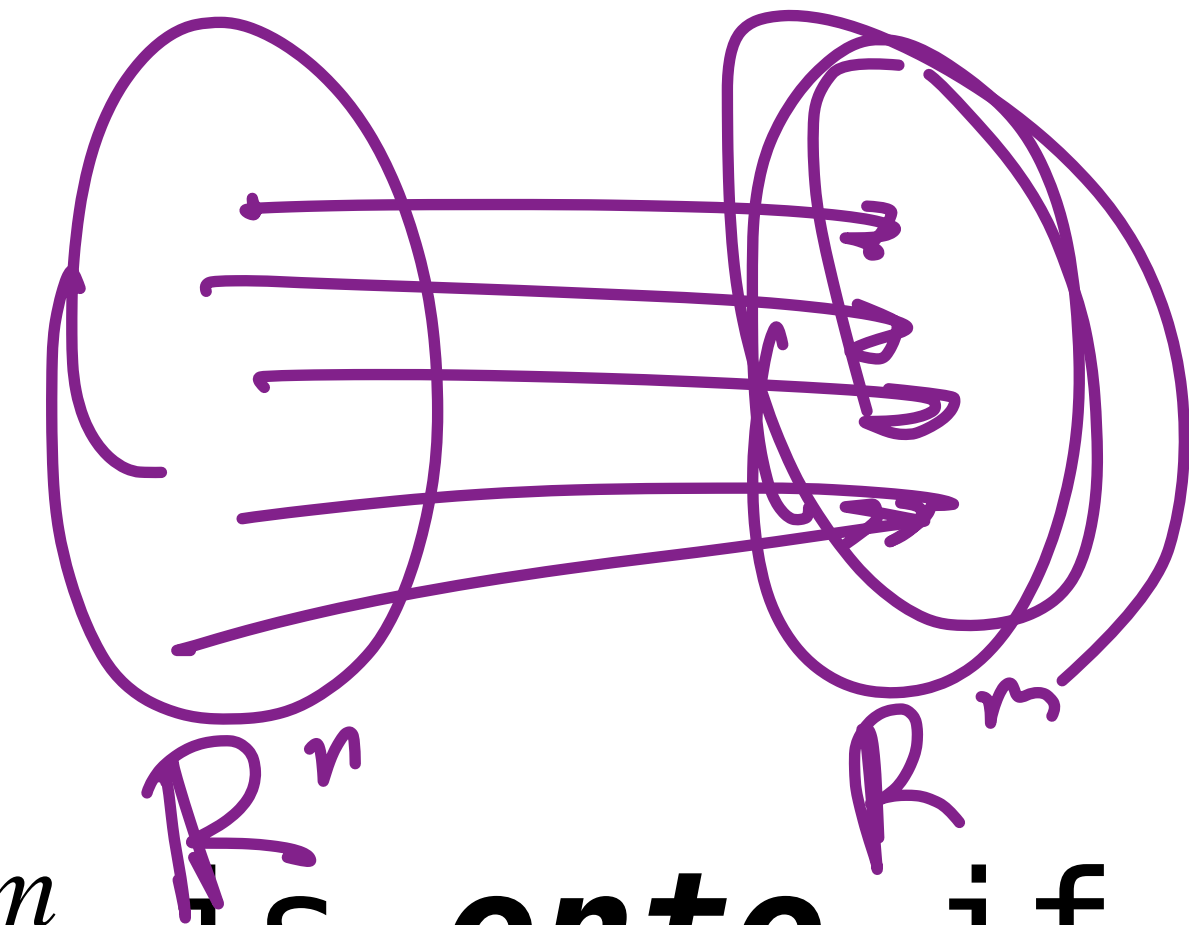
$\mathbf{b} = \vec{0}$ has more than 1
sol.

Onto Transformations

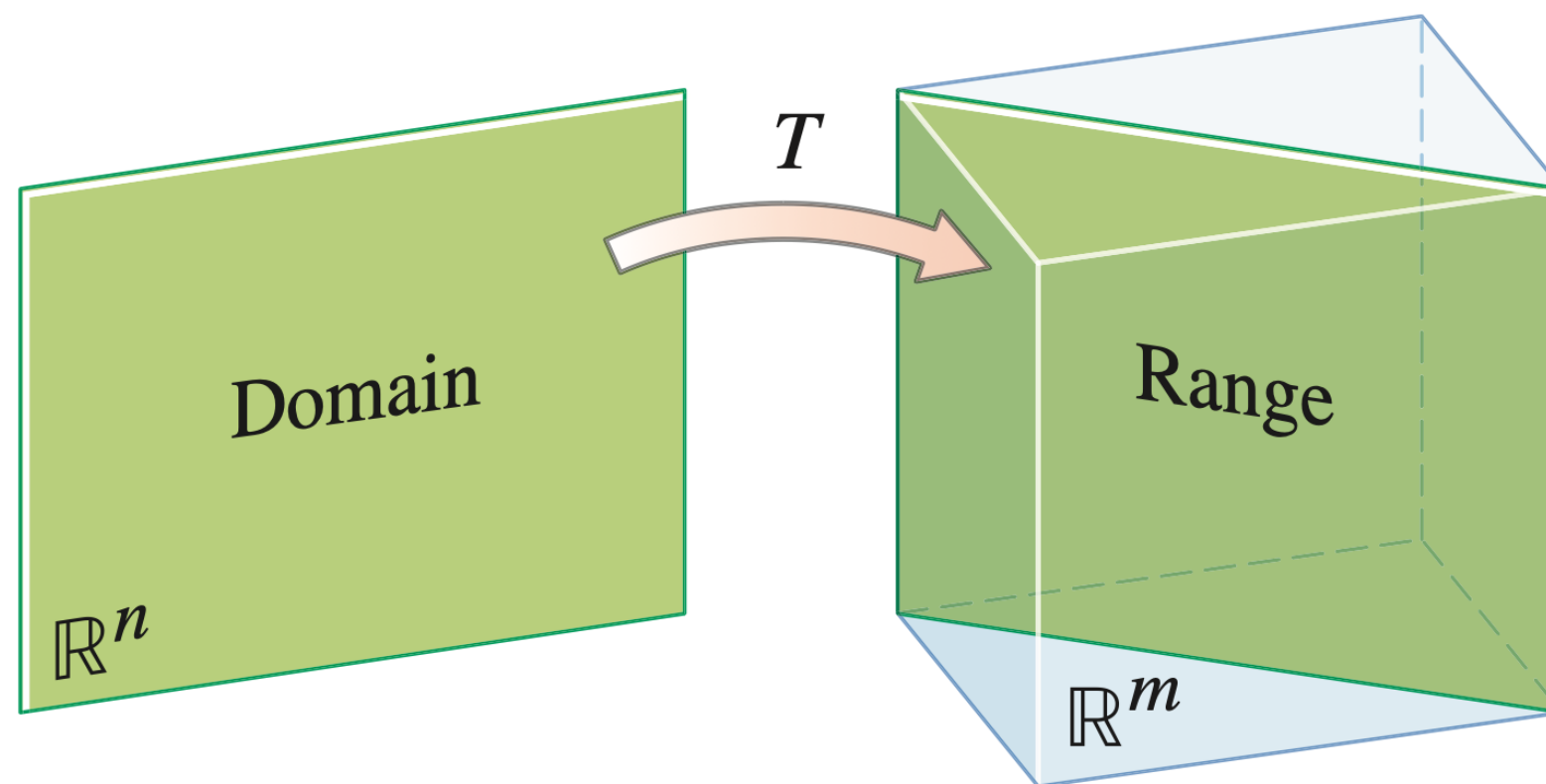
Onto Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is ***onto*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at least one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

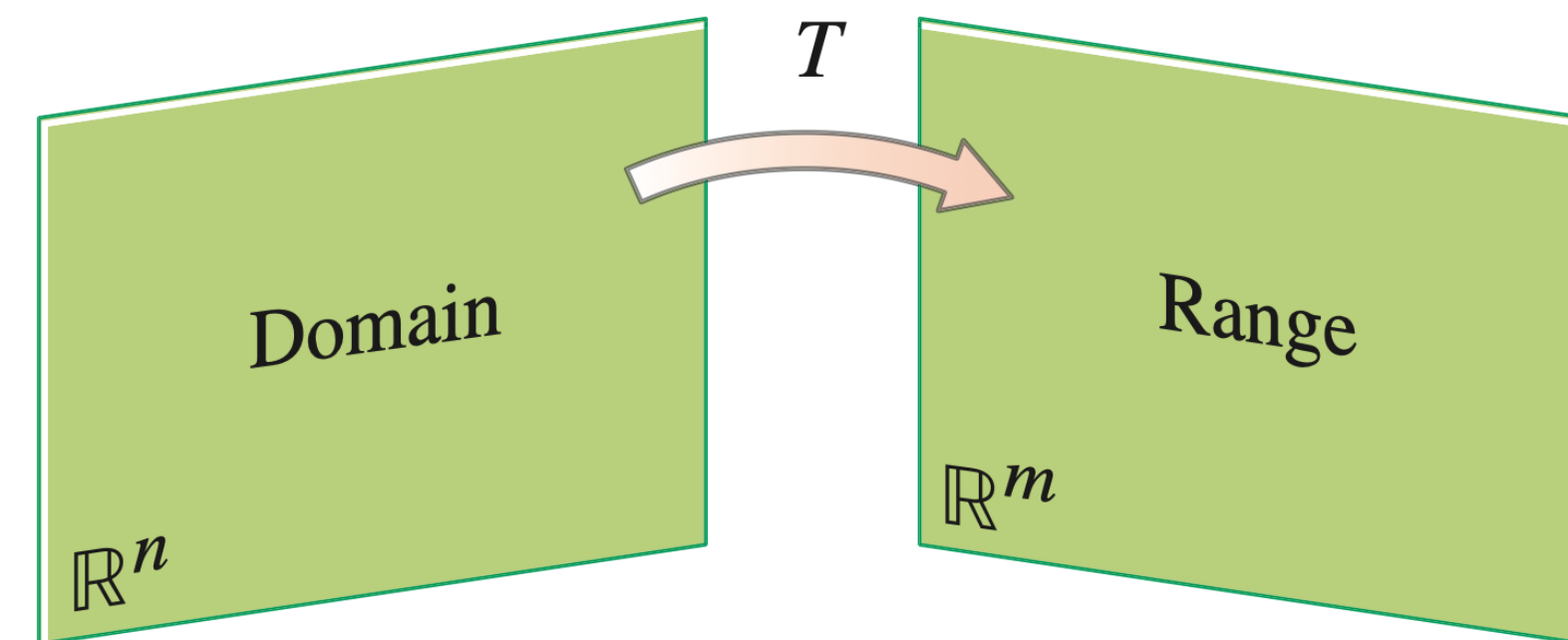
Onto Transformations



Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at least one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)



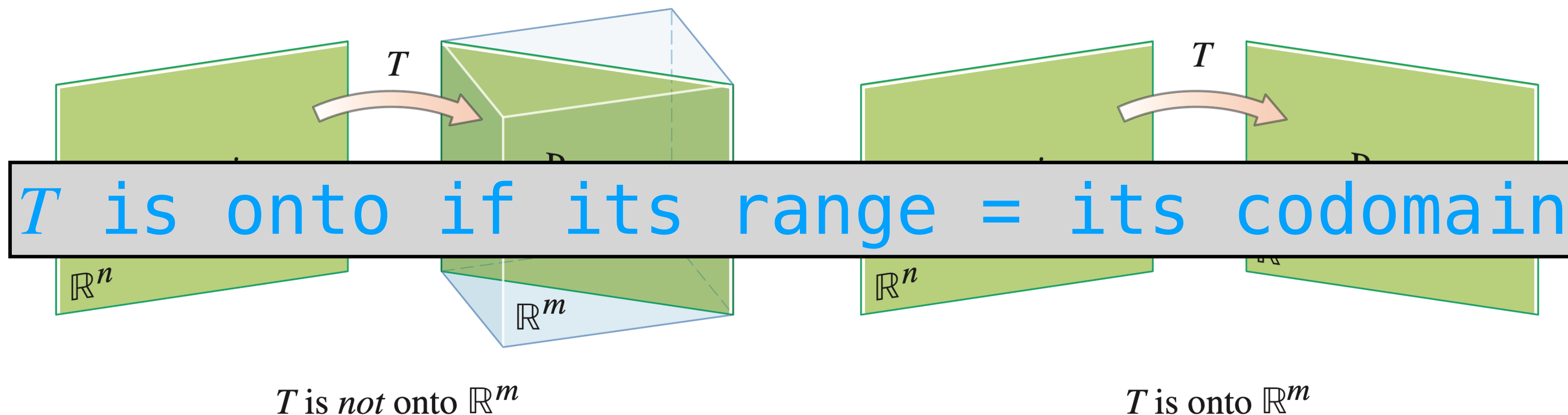
T is *not* onto $\mathbb{R}^m$



T is onto $\mathbb{R}^m$

Onto Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at least one vector $\mathbf{v}$ in $\mathbb{R}^n$** (where $T(\mathbf{v}) = \mathbf{b}$)



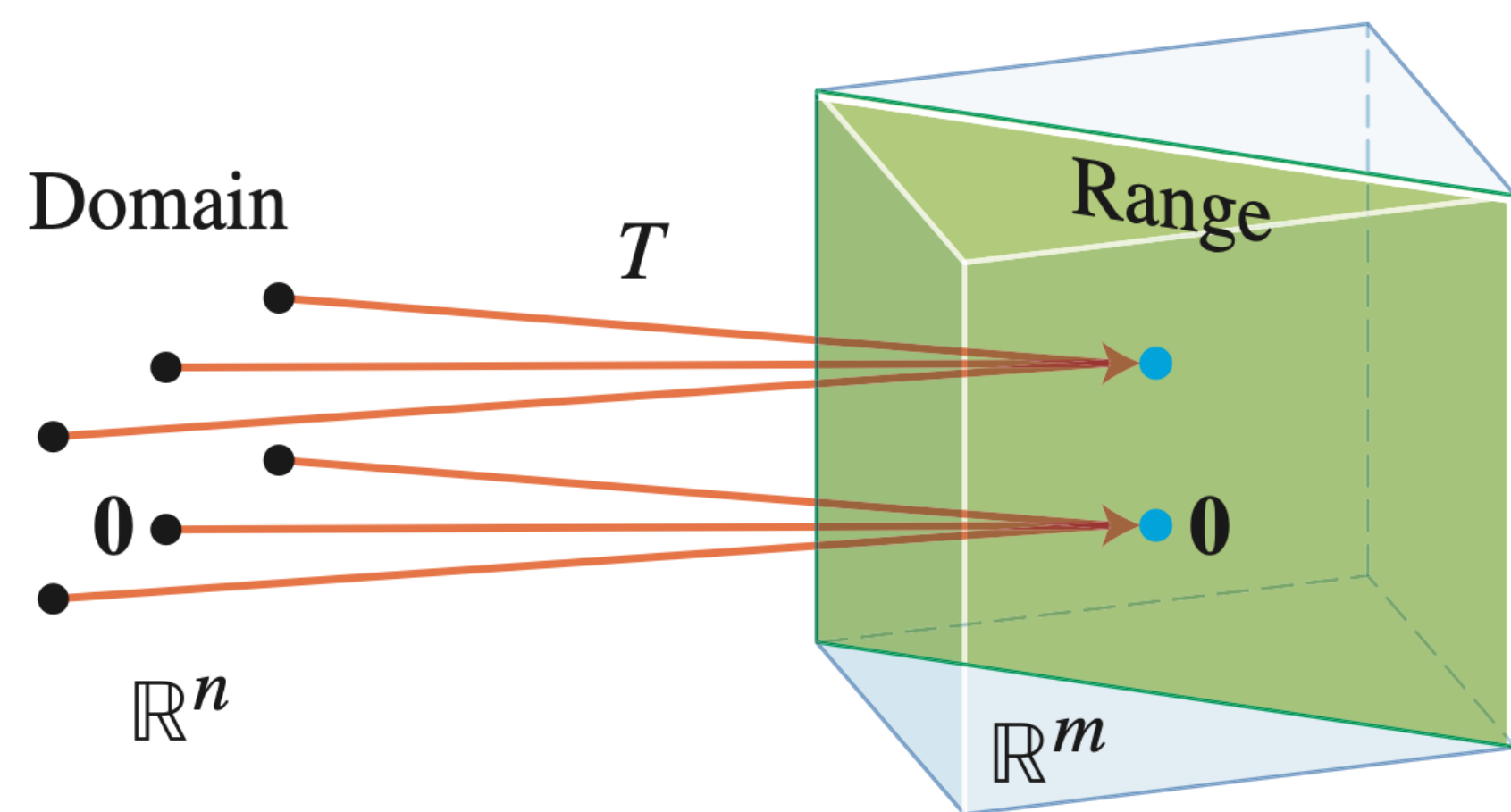
One-to-one Transformations

One-to-one Transformations

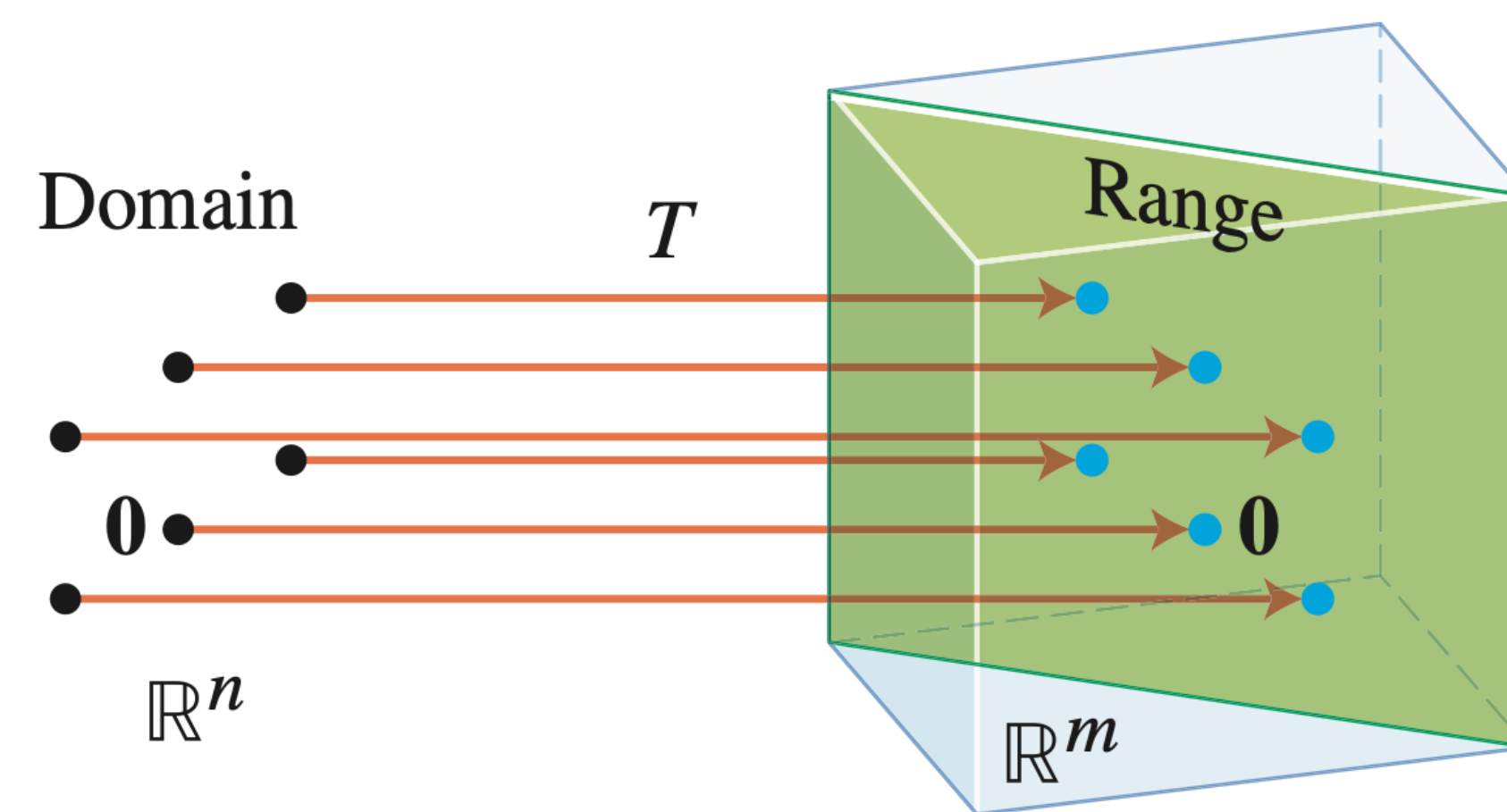
Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is ***one-to-one*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

One-to-one Transformations

Def. A transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)



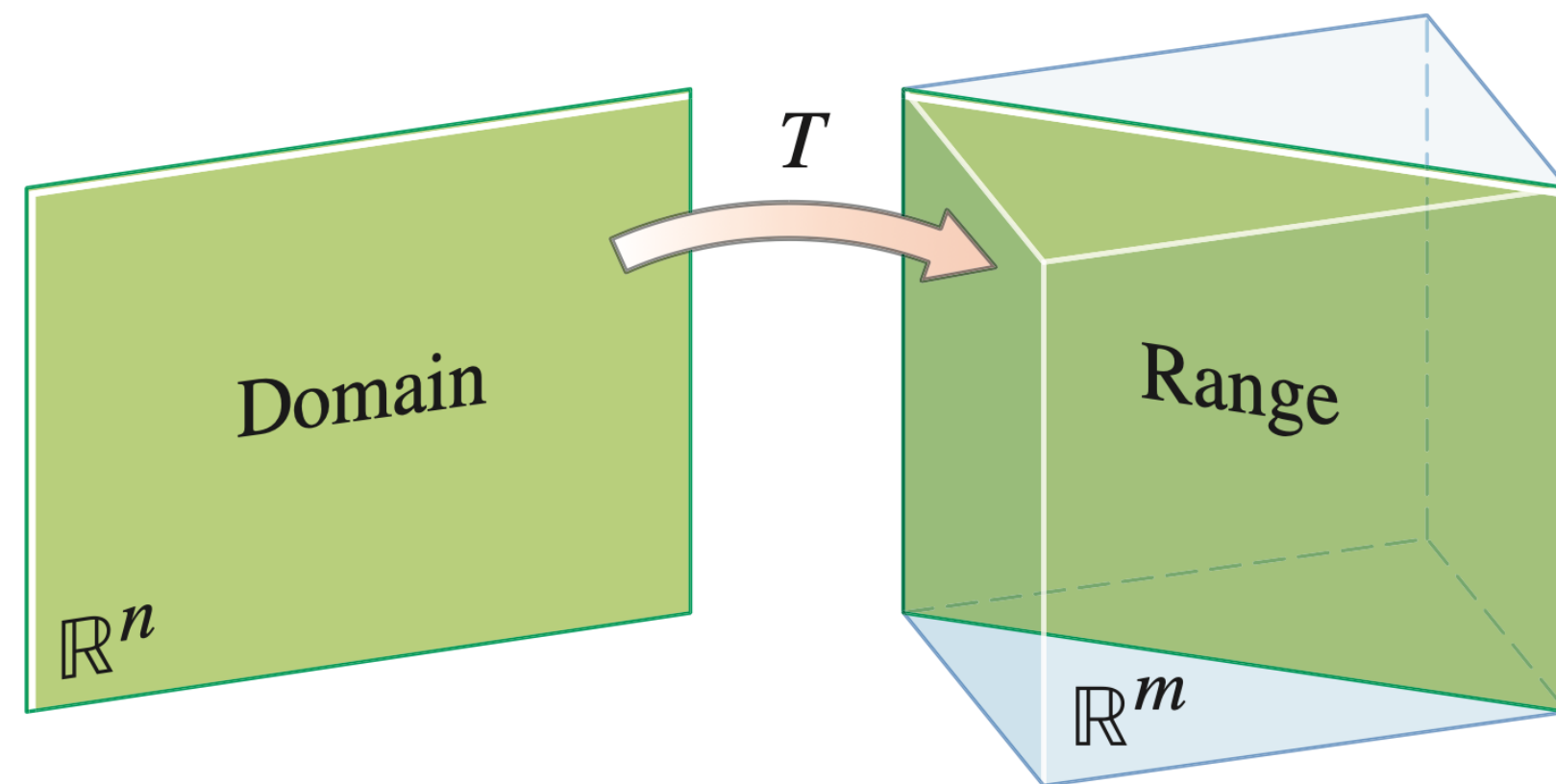
T is *not* one-to-one



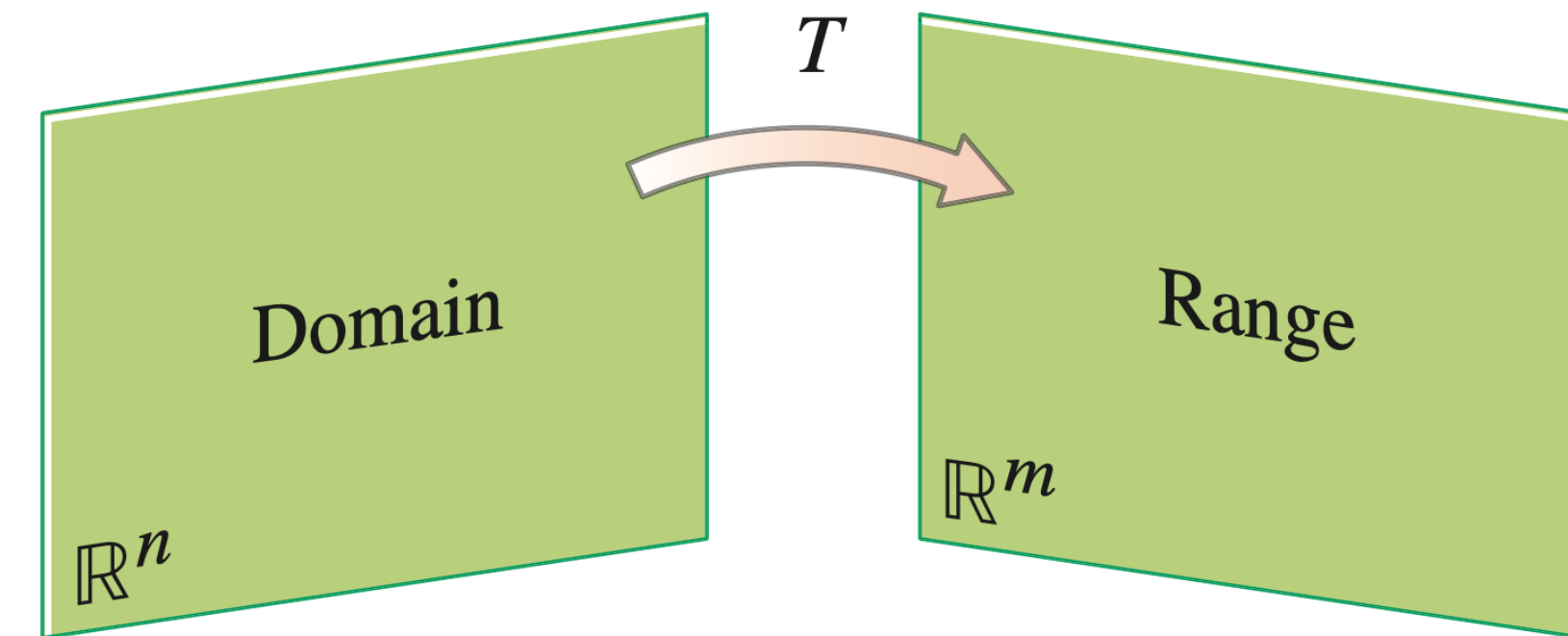
T is one-to-one

(no collision)

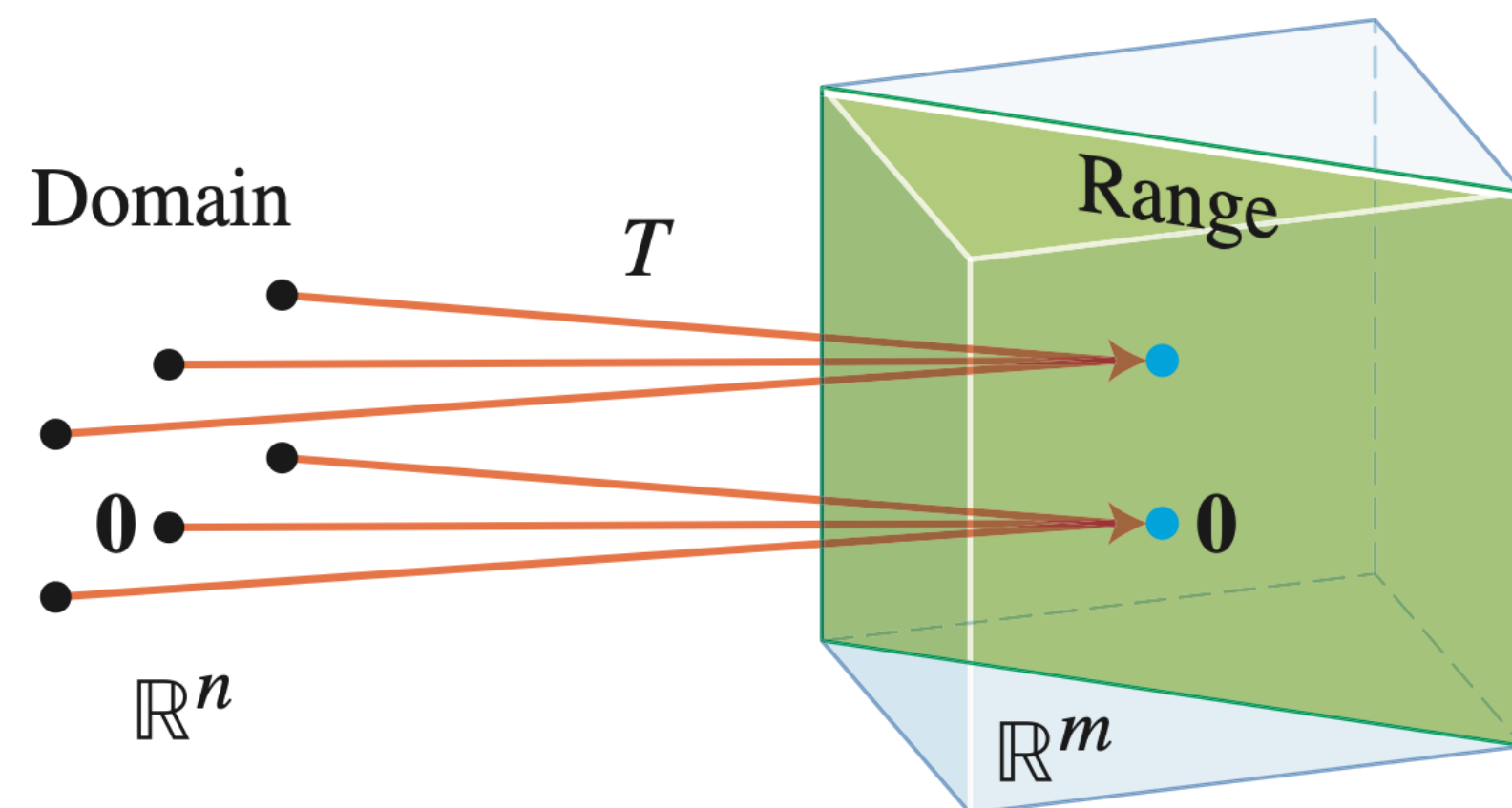
Comparing Pictures



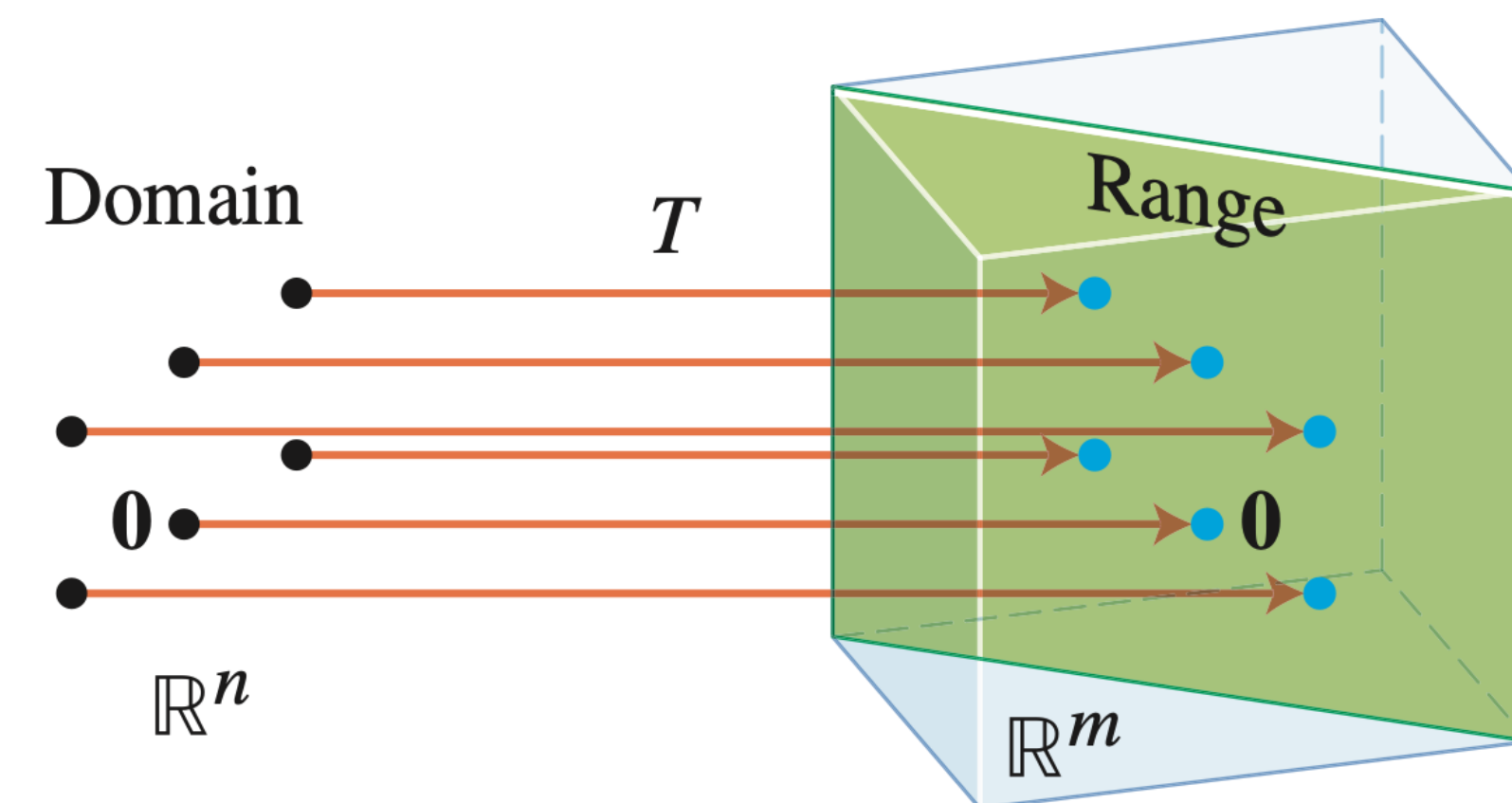
T is not onto $\mathbb{R}^m$



T is onto $\mathbb{R}^m$



T is not one-to-one



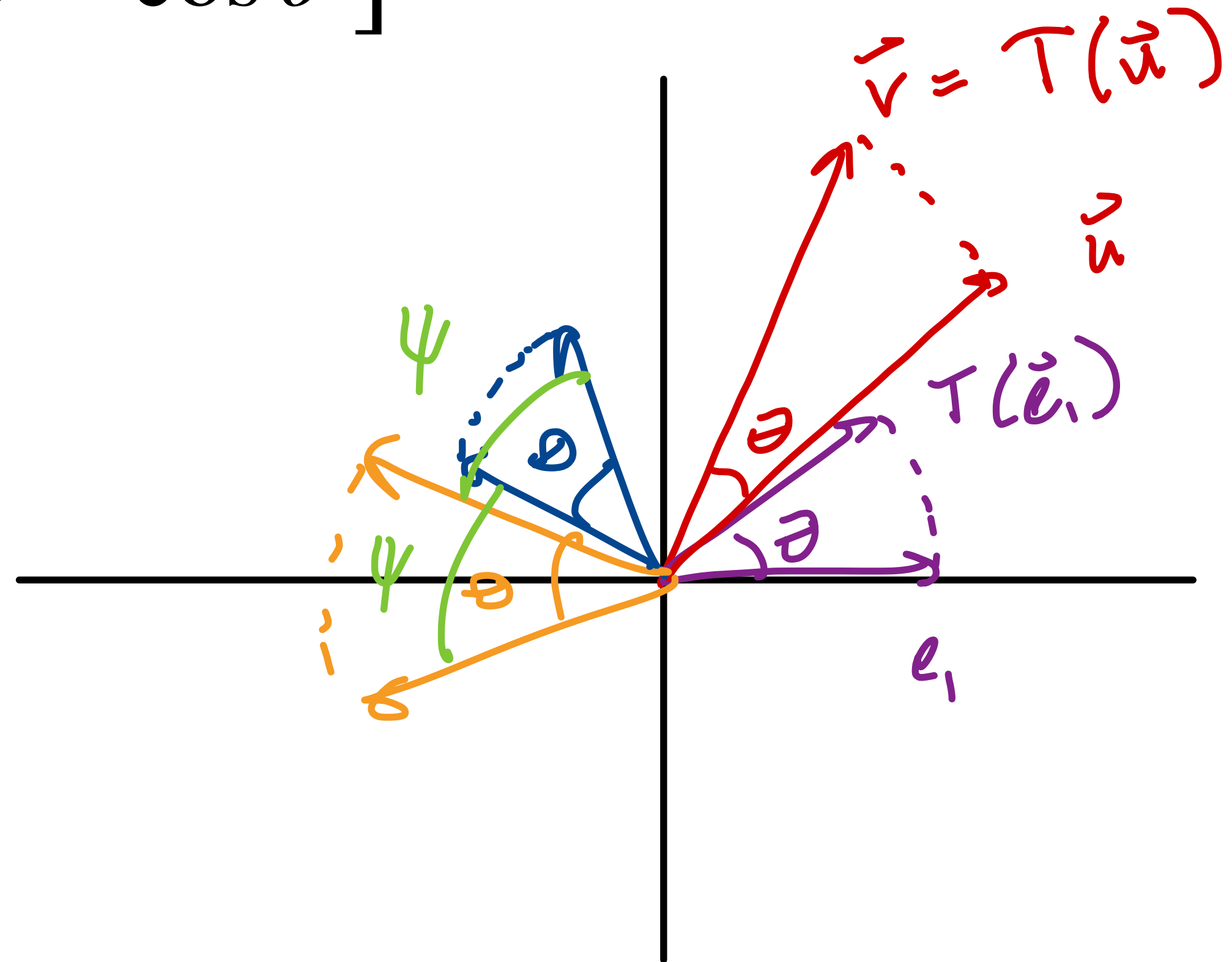
T is one-to-one

Example: 1-1 and Onto

$$\mathbf{x} \mapsto \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \mathbf{x}$$

onto: rotate clockwise

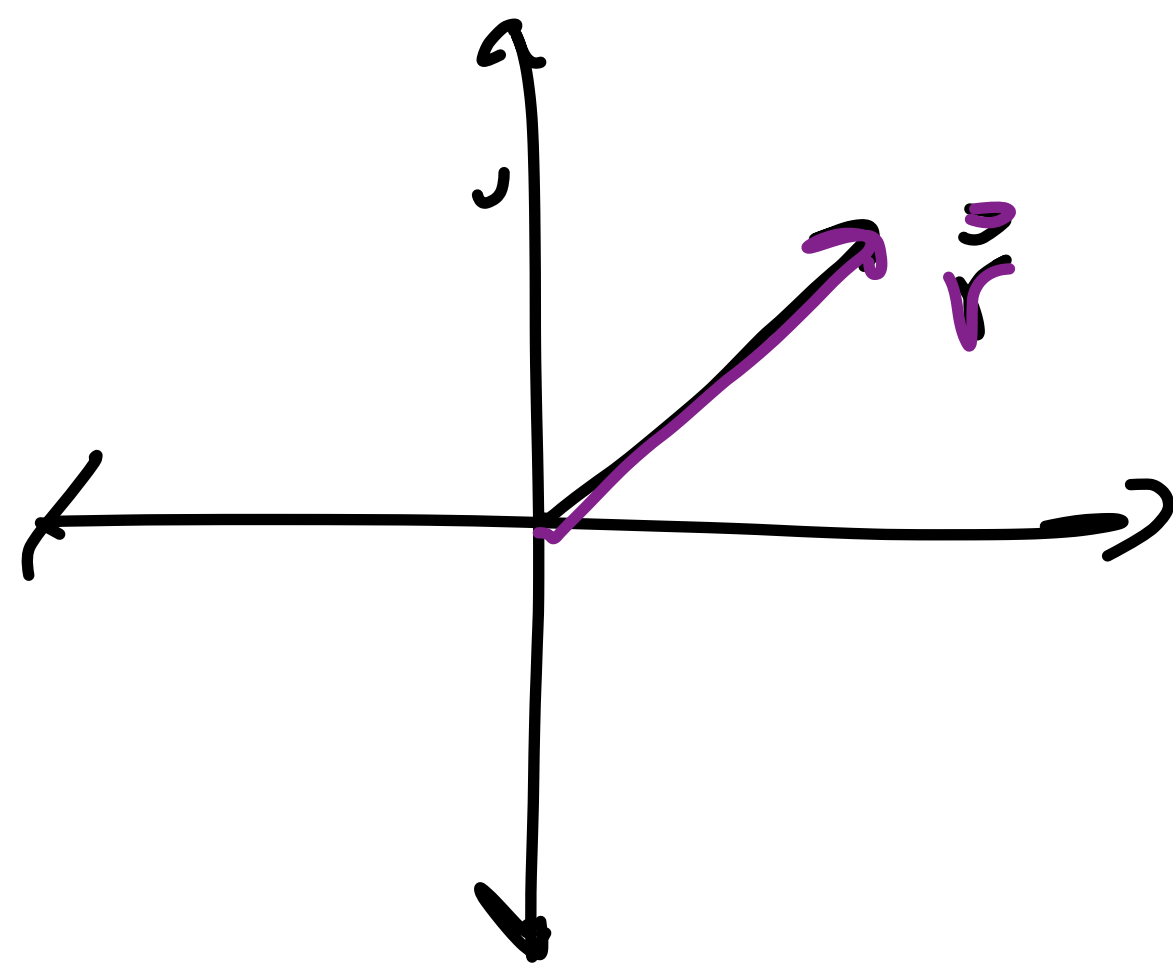
1-1: no collisions



Example: 1-1 and not Onto

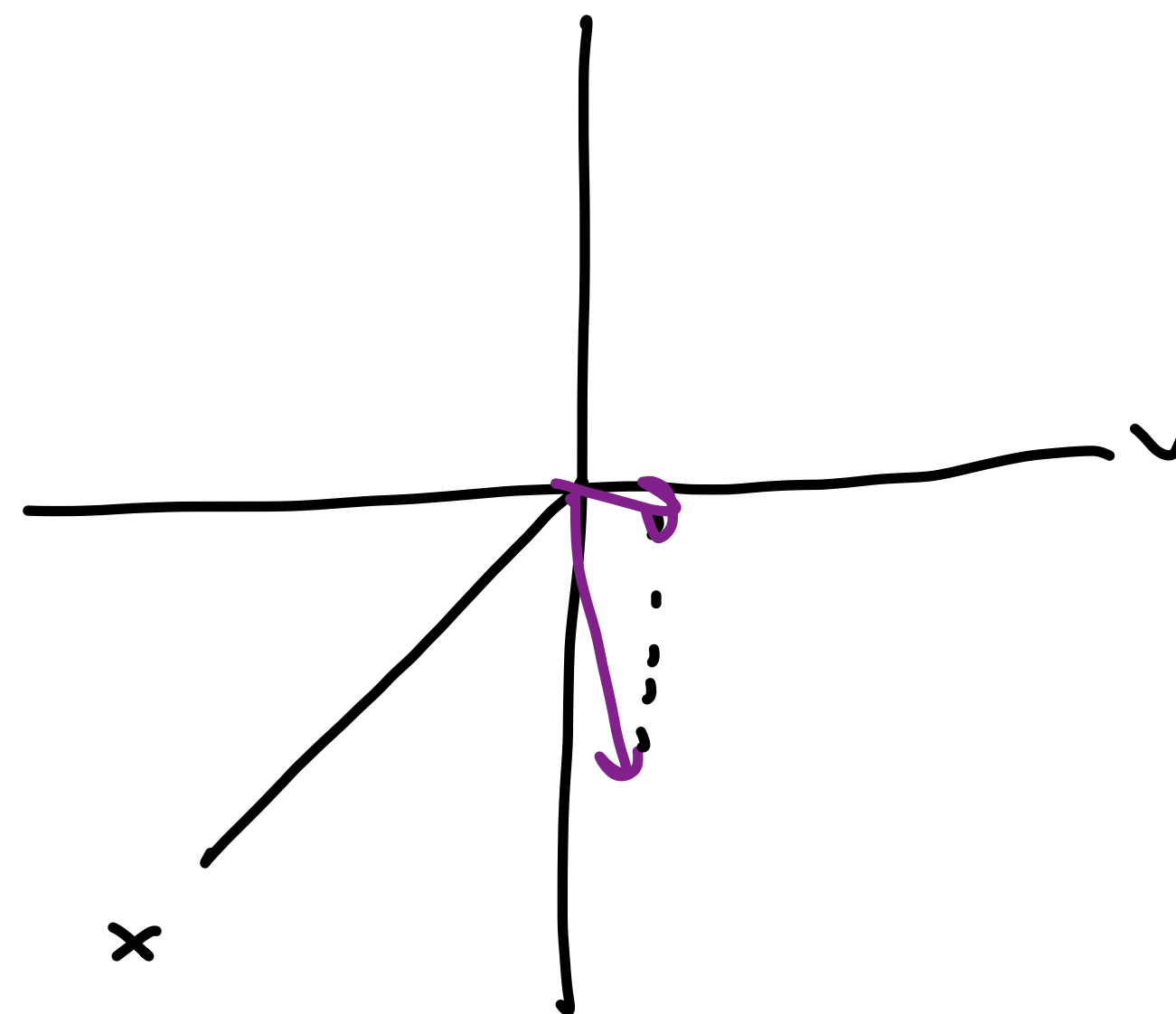
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \notin \text{ran}(T)$$



1-1

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

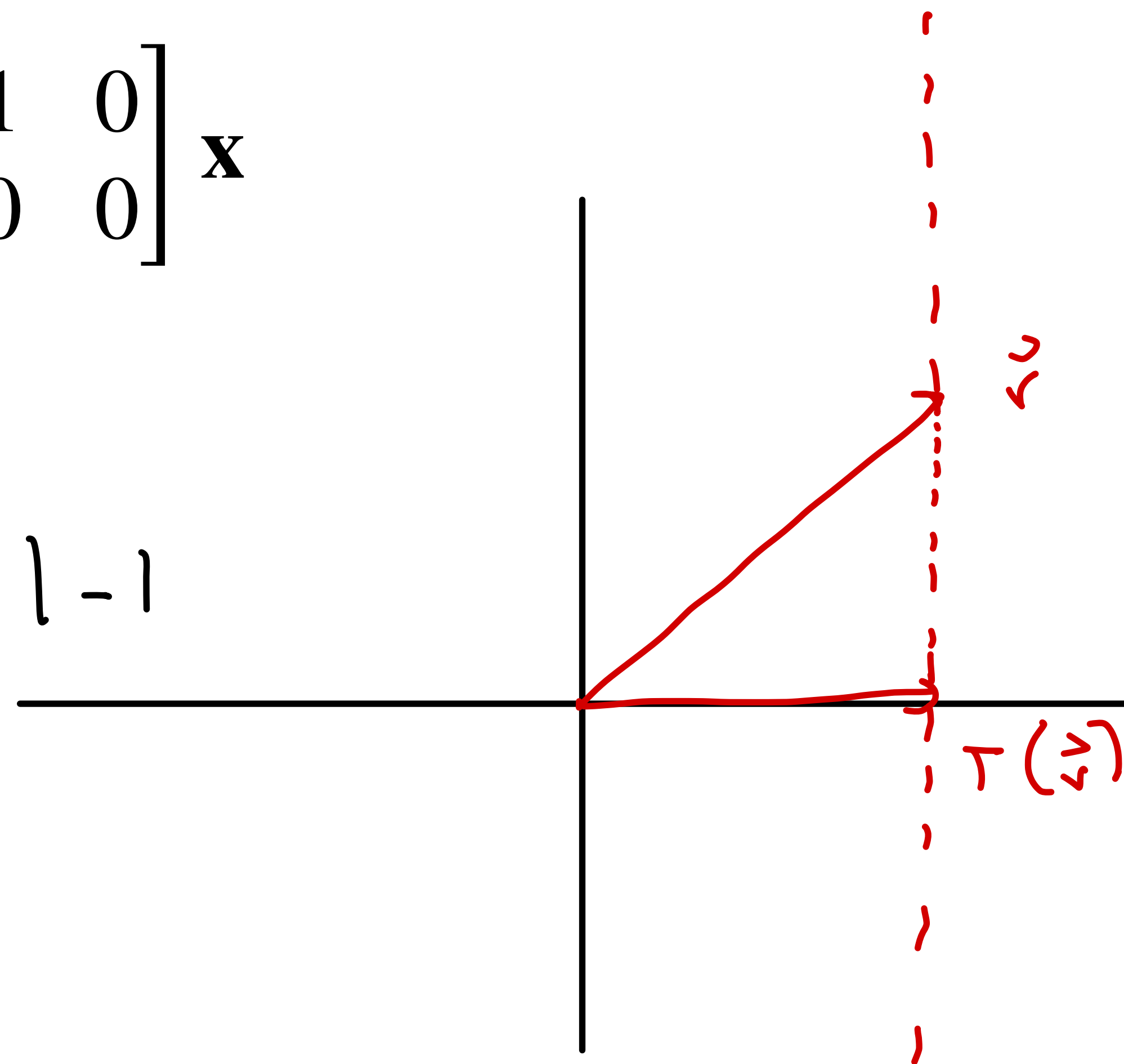


Example: Not 1-1 and not Onto

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin \text{ran}(T), \text{ not onto} \quad \mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}$$

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right), \text{ not 1-1}$$

collision



Example: Not 1-1 and Onto

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathcal{T}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right) = \mathcal{T}\left(\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}\right), \text{ not 1-1}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \begin{bmatrix} 5 \\ 7 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Taking Stock: Onto

Taking Stock: Onto

Thm. The following are equivalent for linear ~~T~~
 $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

Taking Stock: Onto

Thm. The following are equivalent for linear =
 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is onto

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Thm. The following are equivalent for linear =
 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

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» $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$

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 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

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» $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$

» $\text{range}(T) = \text{codomain}(T)$

Taking Stock: Onto

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 $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
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- » the columns of A span $\mathbb{R}^m$

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- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$
- » the columns of A span $\mathbb{R}^m$
- » A has a pivot position in every row

Taking Stock: One-to-One

Taking Stock: One-to-One

Thm. The following are equivalent for linear
 $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

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Taking Stock: One-to-One

Thm. The following are equivalent for linear $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is one-to-one

» $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$

Taking Stock: One-to-One

Thm. The following are equivalent for linear $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

Taking Stock: One-to-One

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Taking Stock: One-to-One

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- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
- » The columns of A are linearly independent
- » A has a pivot position in every column

How To: One-to-One and Onto

Question. Show that the linear transformation T is one-to-one/onto

Solution. (one approach) Find the matrix which implements T and see if it has a pivot in every column/row

Warning: this is not the only way. Always try to think if you can solve it using *any* of the perspectives

Example: 1-1 and Onto

$$\mathbf{x} \mapsto \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \mathbf{x}$$

$\simeq$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example: 1-1 and not Onto

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$$

Example: Not 1-1 and not Onto

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}$$

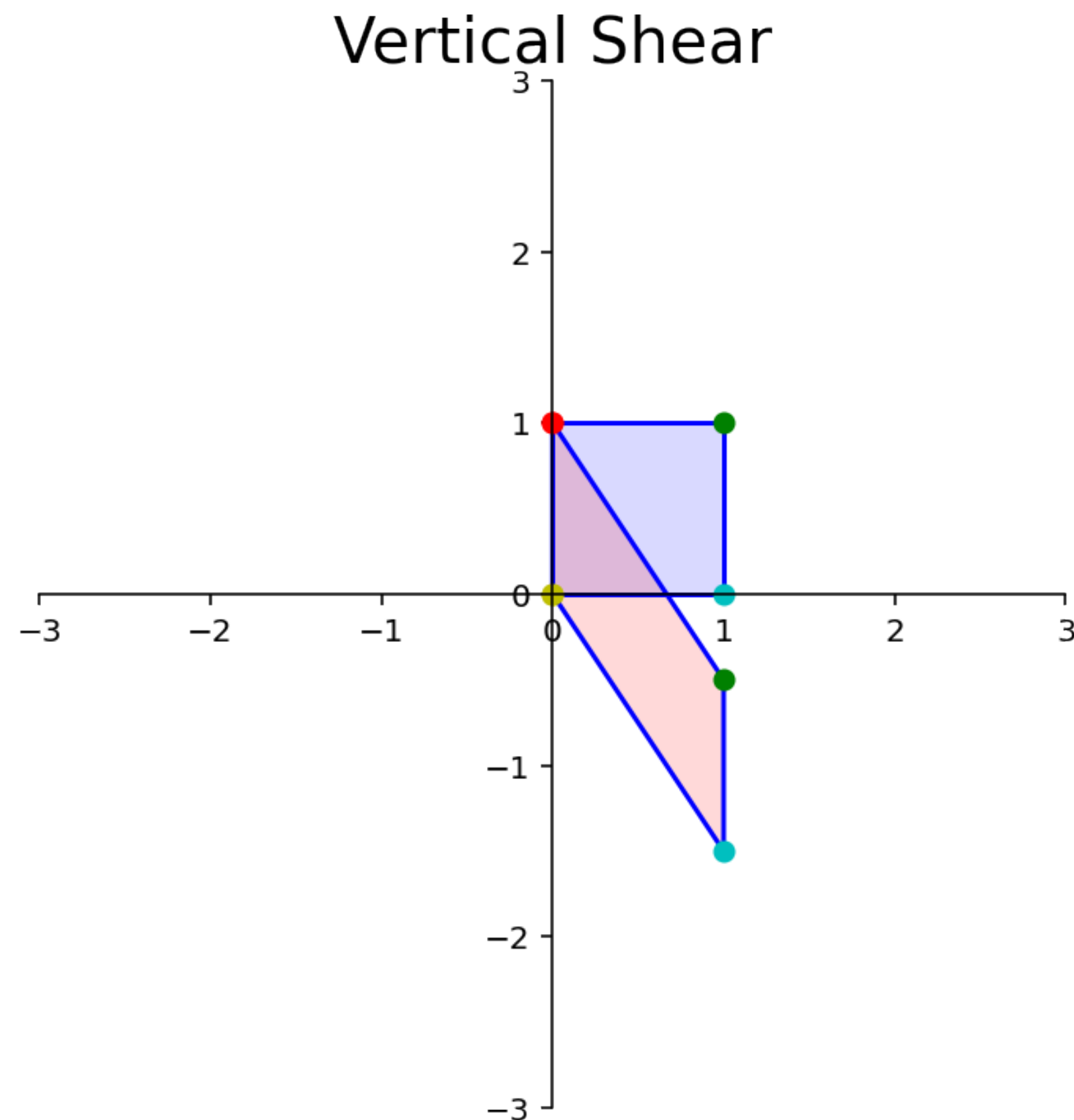
Example: Not 1-1 and Onto

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

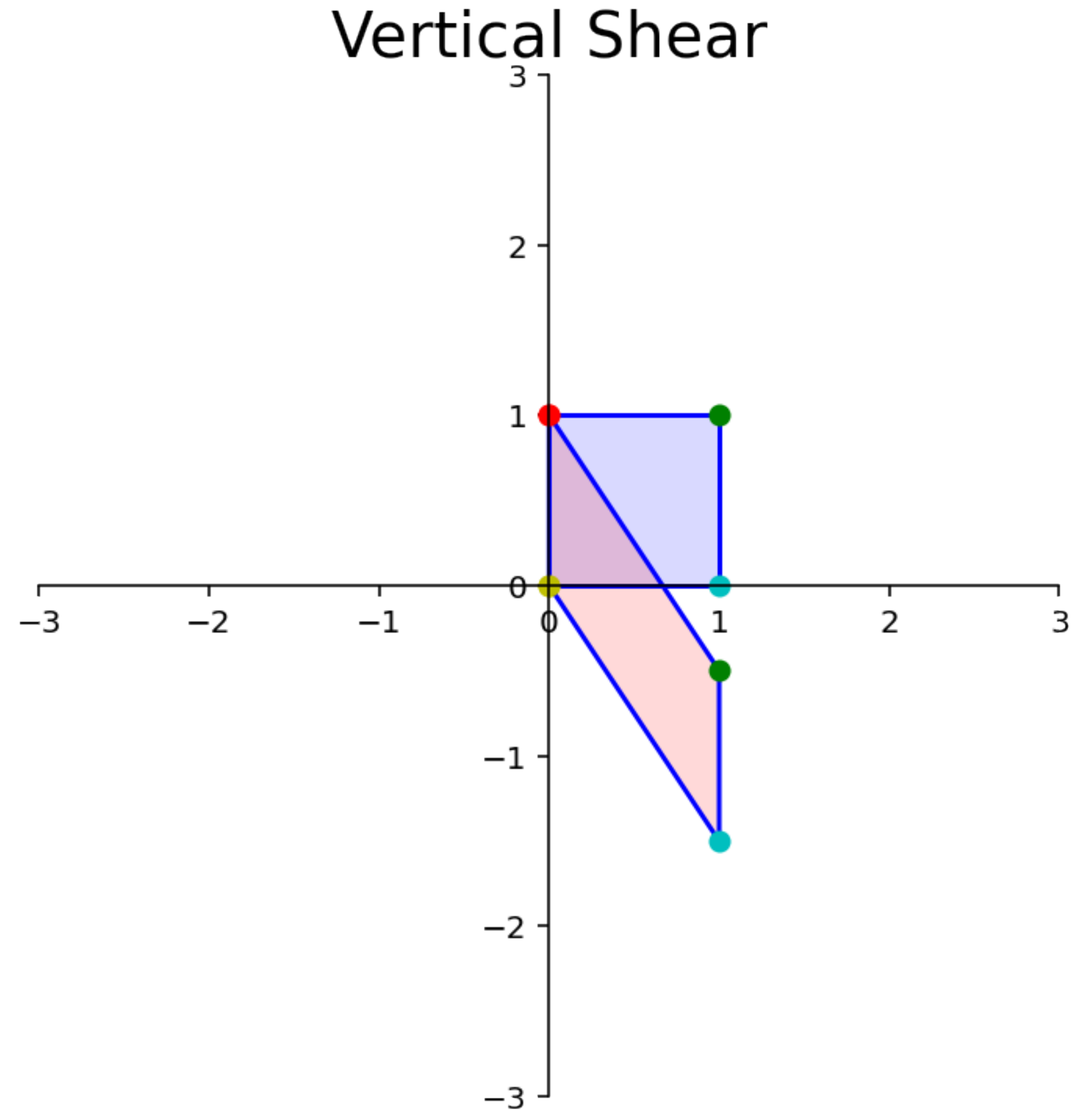
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Question

Is vertical shearing a 1-1 transformation?
Justify your answer

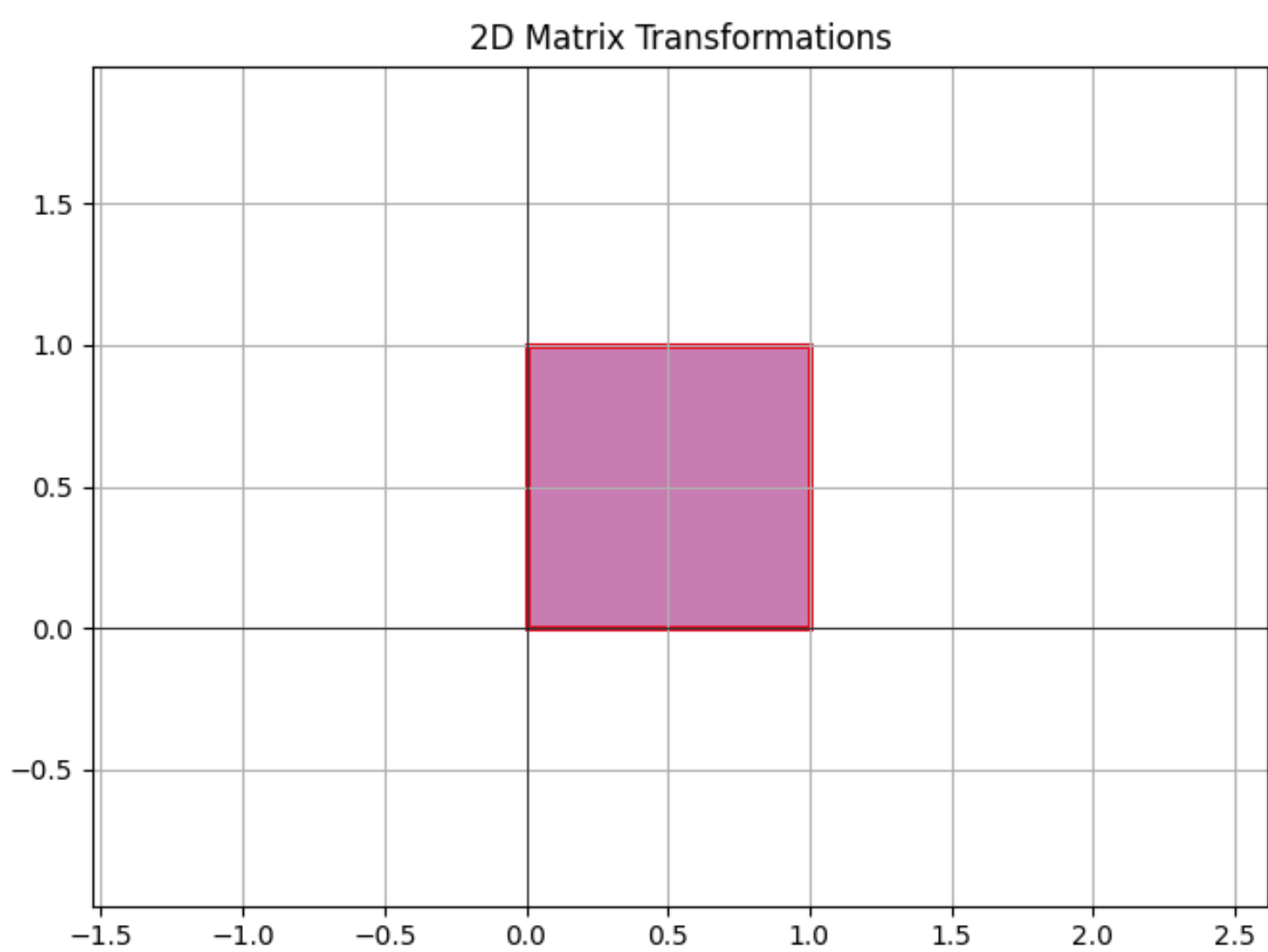


Answer: Yes

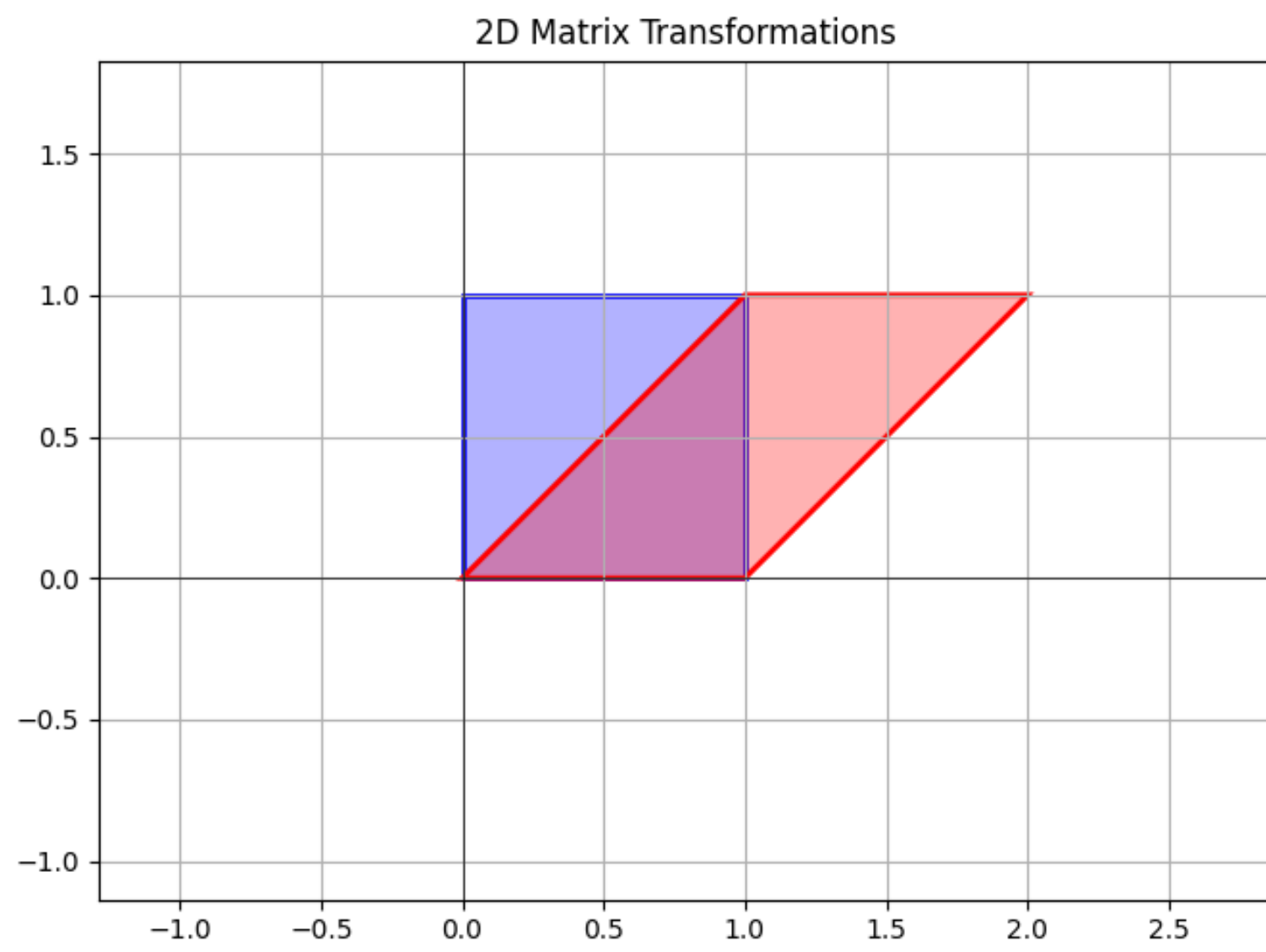


Composing Linear Transformations

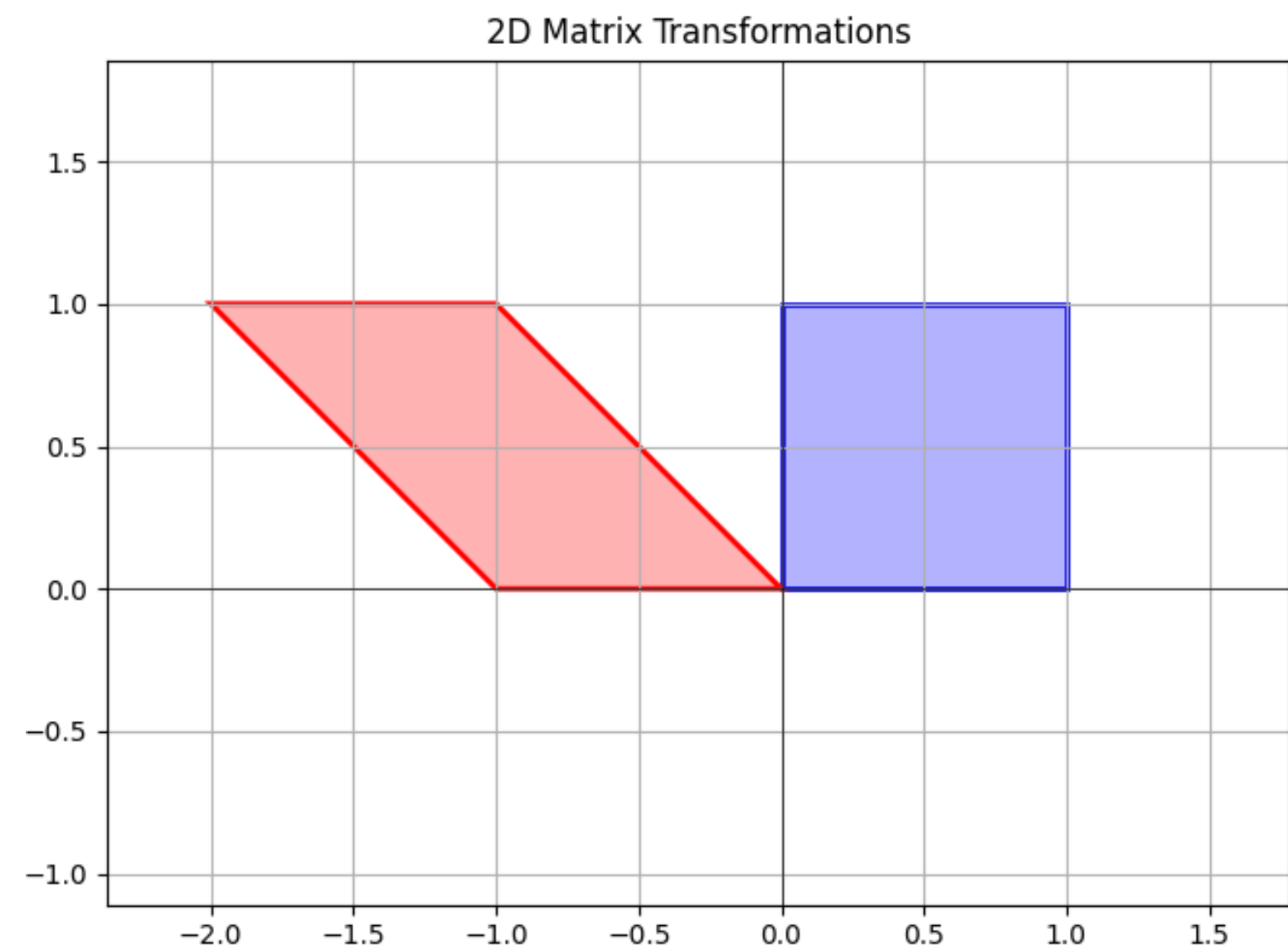
Shearing and Reflecting (Geometrically)



shear



reflect

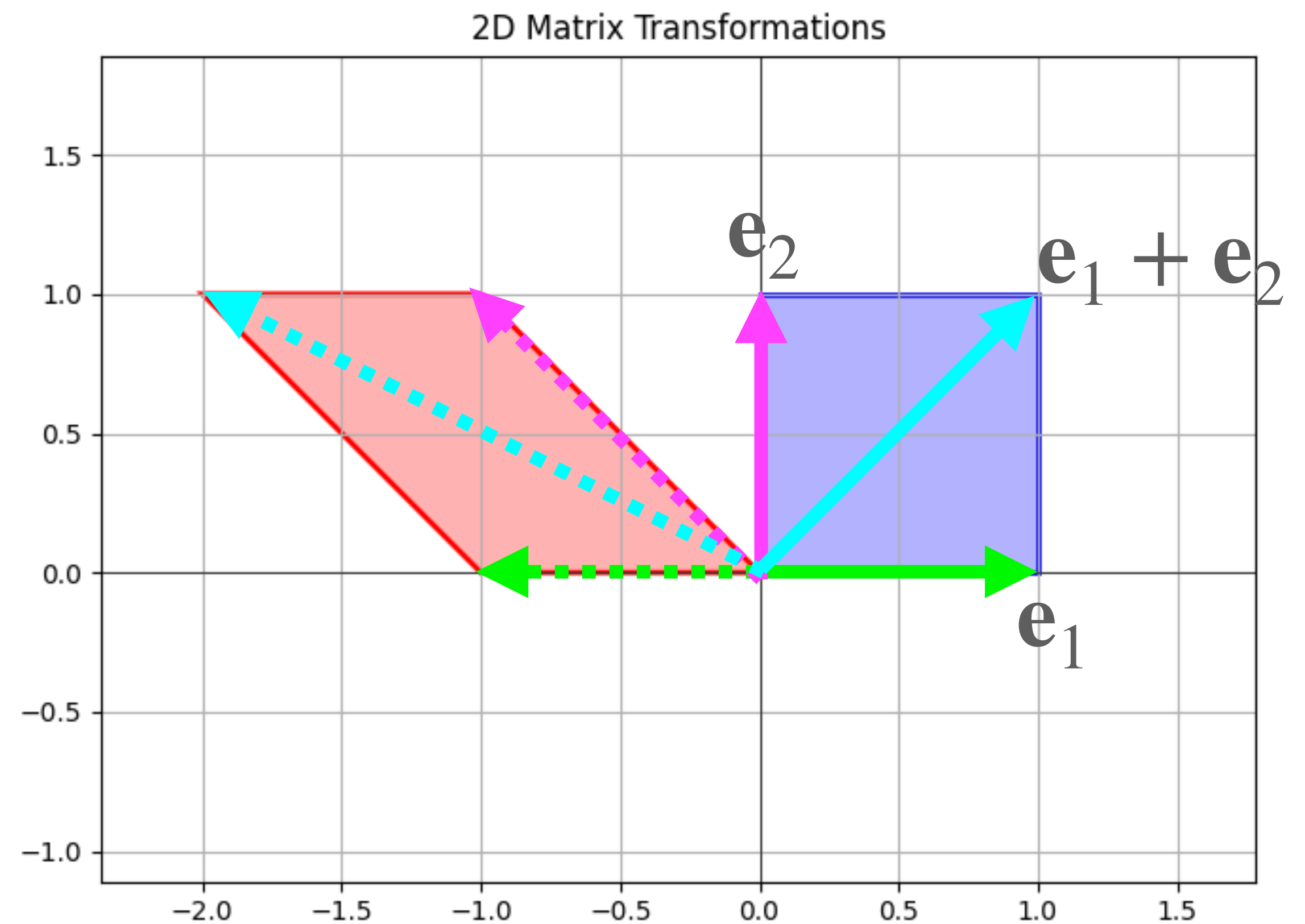


Shearing and Reflecting Matrix

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



Shearing and Reflecting (Algebraically)

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

reflect shear

First multiply by shear matrix, then multiply
by reflection matrix

Shearing and Reflecting (Algebraically)

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

reflect shear

First multiply by shear matrix, then multiply
by reflection matrix

This gives us the same transformation

Shearing and Reflecting

$$\begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x} \right)$$

The Key Fact

The Key Fact

Thm. The composition of two linear transformations is a linear transformation

S, T

$$\begin{aligned} S(T(\vec{v}_1 + \vec{v}_2)) &= S(T(\vec{v}_1) + T(\vec{v}_2)) && \text{(additivity)} \\ &= S(T(\vec{v}_1)) + S(T(\vec{v}_2)) \end{aligned}$$

exercise: show homogeneity.

The Key Fact

Thm. The composition of two linear transformations is a linear transformation

This means the composition of two matrix transformations can be represented as a *single* matrix

The Key Question

*Given two linear transformations,
how to we compute the matrix which
implements their composition?*

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*Given two linear transformations,
how to we compute the matrix which
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Matrix Multiplication

Matrix Multiplication

Shearing and Reflecting

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) =$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) =$$

$$x_1 \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

General Composition (2D)

$$\begin{aligned} A \left(\begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= A \left(x_1 \vec{b}_1 + x_2 \vec{b}_2 \right) \\ &= A(x_1 \vec{b}_1) + A(x_2 \vec{b}_2) \\ &= x_1 A \vec{b}_1 + x_2 A \vec{b}_2 = \boxed{\begin{bmatrix} A \vec{b}_1 & A \vec{b}_2 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

Matrix Multiplication

Def. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ with columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$ the product $AB \in \mathbb{R}^{m \times p}$ is given by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix}$$

Replace each column of B with A multiplied by that column

Tracking Dimensions

This only works if the number of columns of the left matrix matches the number of rows of the right matrix

The diagram illustrates matrix multiplication with dimensions. On the left, a matrix of size $m \times n$ is shown with 5 rows and 3 columns. A blue vertical line to its left is labeled m . A red horizontal line above it is labeled n . A red vertical line to its right is labeled n . This matrix is multiplied by a second matrix of size $n \times k$ with 3 rows and 4 columns. A purple horizontal line above the second matrix is labeled k . The result is a matrix of size $m \times k$ with 5 rows and 4 columns. A blue vertical line to its left is labeled m . A purple horizontal line above it is labeled k .

$$\begin{matrix} (m \times n) \\ \textcircled{5} \times \textcircled{3} \end{matrix}$$

$$\begin{matrix} (n \times k) \\ \textcircled{3} \times \textcircled{4} \end{matrix}$$

$$\begin{matrix} (m \times k) \\ \textcircled{5} \times \textcircled{4} \end{matrix}$$

Important

Even if AB is defined, it may
be that BA is *not* defined

Non-Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

2×3 2×2

↖ ↗

Non-Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

Handwritten annotations above the equation:

- A red circle around the 3 in the first matrix, with a red arrow pointing to the 2×3 dimension label.
- A red circle around the 2 in the second matrix, with a red arrow pointing to the $\mathbb{R}^2$ dimension label.

Large red 'X' marks are placed over the two terms in the brackets on the right side of the equation.

These are not defined

Example

$$\begin{array}{c} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \\ \begin{array}{cc} 2 \times 2 & 2 \times 3 \end{array} \end{array} = \begin{array}{c} \begin{array}{cc} 2 \times 2 & \mathbb{R}^2 \end{array} \\ \begin{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{bmatrix} \\ \begin{array}{c} 2 \times 3 \end{array} \end{array}$$

The Key Fact (Restated)

For any matrices A and B such that AB is defined and any vector $\mathbf{v}$

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

The matrix implementing the composition is the product of the two underlying matrices

Row-Column Rule

Given $A \in \mathbb{R}^{m \times n}$ and $n \times p$ matrix $B \in \mathbb{R}^{n \times p}$ the entry $(AB)_{ij}$ is given as:

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

dot product

Example

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} -1(1) + 0(0) & -1(1) + 0(1) \\ 0(1) + 1(0) & 0(1) + 1(1) \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its first column highlighted in light red. An equals sign follows, and then a 5x4 matrix where its first row is highlighted in light purple. All elements in the matrices are represented by asterisks (*).

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

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$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

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A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its third column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element at the intersection of the first row and third column is highlighted in light purple, representing the result of the dot product of the first row of the first matrix and the third column of the second matrix.

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element in the first row and fourth column is highlighted in light purple, representing the result of the dot product of the first row of the first matrix and the fourth column of the second matrix.

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements represented by asterisks. The third matrix is a 5x4 matrix, also with all elements represented by asterisks. An equals sign (=) is placed between the second and third matrices. The first matrix has its second row highlighted in light blue. The second matrix has its first column highlighted in light red. The third matrix has its first row highlighted in light purple. This highlights the specific row and column used in the calculation of a single element in the product matrix.

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements as asterisks, but its second column is highlighted with a light red background. An equals sign (=) follows. The third matrix is a 5x4 matrix with all elements as asterisks, but its second column is highlighted with a light purple background. This illustrates that the second row of the first matrix multiplied by the second column of the second matrix results in the second column of the third matrix.

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Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements represented by asterisks (*). The third matrix is a 5x4 matrix, also with all elements represented by asterisks (*). The first matrix has its second row highlighted in light blue. The second matrix has its third column highlighted in light red. The third matrix has its third column highlighted in light purple. An equals sign (=) is placed between the second and third matrices, indicating that the product of the first two matrices is the third matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all cells containing an asterisk (*). The second matrix is a 3x4 matrix, also with all cells containing an asterisk (*). The third matrix is a 5x4 matrix, also with all cells containing an asterisk (*). The second matrix is highlighted with a light red background, and the third matrix is highlighted with a light purple background. The first matrix is highlighted with a light blue background. The second and third matrices are separated by an equals sign (=).

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Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with a light blue horizontal band highlighting its third row. The second matrix is a 3x4 matrix with a light red vertical band highlighting its second column. An equals sign follows, and then a 5x4 matrix where the element at the intersection of the third row and second column is highlighted with a light purple square, representing the result of the dot product of the third row of the first matrix and the second column of the second matrix.

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 grid of asterisks with its third row highlighted in light blue. The second matrix is a 3x4 grid of asterisks with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 grid of asterisks where the element at the intersection of the third row and fourth column is highlighted in light purple.

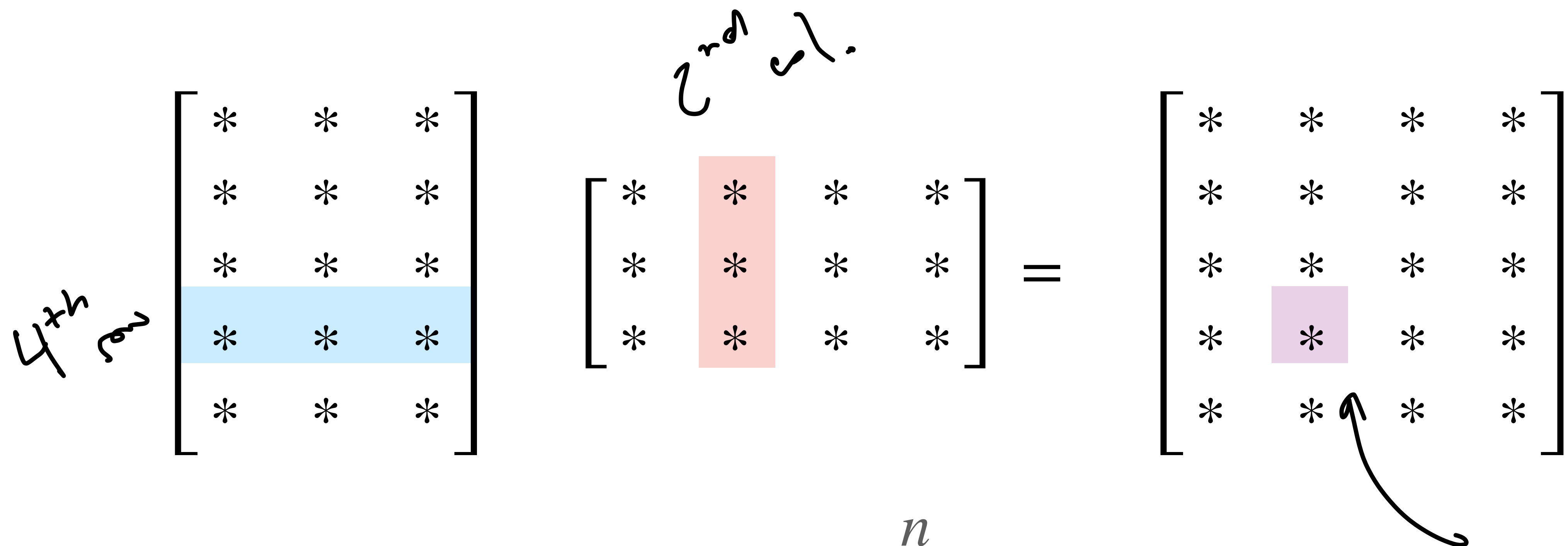
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The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the fourth row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks; the first column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the element in the fourth row and first column is highlighted in light purple, representing the dot product of the fourth row of the first matrix and the first column of the second matrix.

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Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with a light blue horizontal band highlighting its fourth row. The second matrix is a 4x4 matrix with a light red vertical band highlighting its fourth column. An equals sign follows, and then a 5x4 matrix with a light purple square highlighting the element at the intersection of its fourth row and fourth column. All elements in the matrices are represented by asterisks.

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Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the bottom row is highlighted with a light blue background. The second matrix is a 3x4 matrix with asterisks; the first column is highlighted with a light red background. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the bottom-left cell is highlighted with a light purple background, representing the dot product of the bottom row of the first matrix and the first column of the second matrix.

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$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Example

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} =$$

Matrix Operations

Matrix-Vector Multiplication

Matrix-Vector Multiplication

What if right matrix is a single column?

Matrix-Vector Multiplication

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$$A[\mathbf{b}_1] = [A\mathbf{b}_1] = A\mathbf{b}_1$$

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This is just vector multiplication

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$$A[\mathbf{b}_1] = [A\mathbf{b}_1] = A\mathbf{b}_1$$

This is just vector multiplication

We can think of $[A\mathbf{b}_1 \ A\mathbf{b}_2 \ \dots \ A\mathbf{b}_p]$ as collection of simultaneous matrix-vector multiplications

Matrix "Interface"

multiplication

what does AB mean when A and B are matrices?

addition

what does $A + B$ mean when A and B are matrices?

scaling

what does cA mean when A is matrix and c is a real number?

Matrix "Interface"

multiplication

what does AB mean when A and B are matrices?

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what does $A + B$ mean when A and B are matrices?

scaling

what does cA mean when A is matrix and c is a real number?

These should be consistent with matrix-vector interface and vector interface

Matrix Addition

$$[\mathbf{a}_1 \quad \dots \quad \mathbf{a}_n] + [\mathbf{b}_1 \dots \quad \mathbf{b}_n] = [(\mathbf{a}_1 + \mathbf{b}_1) \quad \dots \quad (\mathbf{a}_n + \mathbf{b}_n)]$$

Addition is done column-wise (or element-wise)

$$\text{e.g. } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} (1+2) & (2+3) \\ (3-2) & (4-3) \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 1 \end{bmatrix}$$

Matrix Addition

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This is exactly the same as vector addition, but for matrices

Matrix Addition and Scaling

$$c [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] = [c\mathbf{a}_1 \quad c\mathbf{a}_2 \quad \dots \quad c\mathbf{a}_n]$$

Scaling and adding happen element-wise (or, equivalently, column-wise)

$$\text{e.g. } 2 \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 2(1) & 2(2) \\ 2(-1) & 2(3) \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ -2 & 6 \end{bmatrix}$$

Matrix Addition and Scaling

$$c [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] = [c\mathbf{a}_1 \quad c\mathbf{a}_2 \quad \dots \quad c\mathbf{a}_n]$$

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This is exactly the same as vector scaling, but for matrices

Algebraic Properties (Addition and Scaling)

$$A + B = B + A$$

comm.

$$(A + B) + C = A + (B + C)$$

assoc.

$$A + 0 = A$$

$$r(A + B) = rA + rB$$

$$(r + s)A = rA + sA$$

$$r(sA) = (rs)A$$

distⁿb.

In these properties A , B , and C are matrices of the same size and r and s are scalars ($\mathbb{R}$)

We need to know/memorize these

Algebraic Properties (Addition and Scaling)

$$A(BC) = (AB)C$$

assoc.

$$A(B + C) = AB + AC$$

$$(B + C)A = BA + CA$$

dist.

In these properties A , B , and C are matrices of the appropriate size so that everything is defined, and r is a scalar

We need to know/memorize these

$$r(AB) = (rA)B = A(rB)$$

$$I_m A = A = A I_n$$

is a matrix.

Matrix Multiplication is not Commutative

Important. AB may not be the same as BA

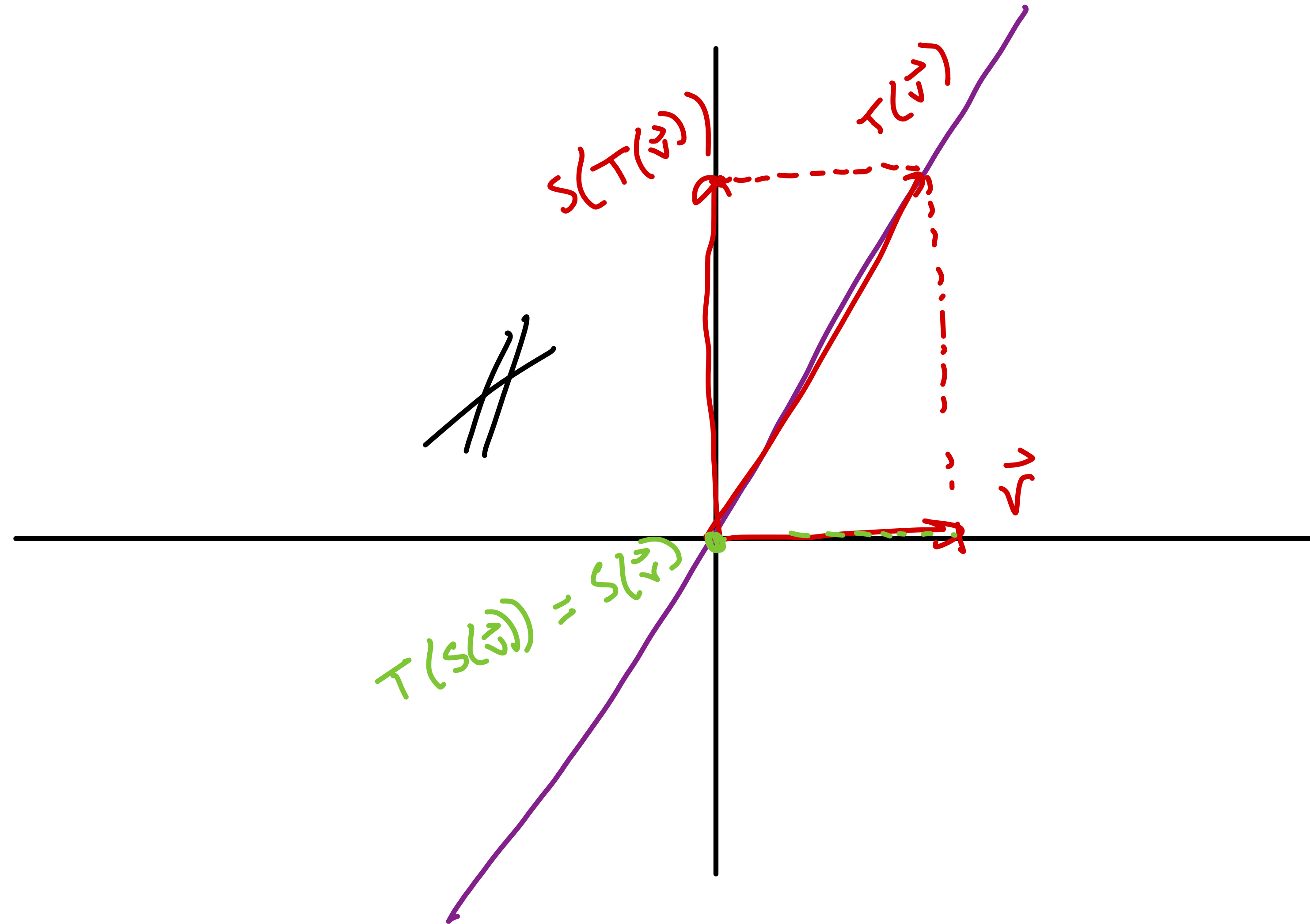
(it may not even be defined)

Question (Conceptual)

Find a pair of 2D linear transformations T_1 and T_2 such that T_1 followed by T_2 is not the same as T_2 followed by T_1

(also find a pair where they are the same)

Answer: Rotation and Reflection



Computational Aspects of Matrix Multiplication

Matrix Operations in Numpy

Let `a` and `b` be 2D numpy arrays and let `c` be a floating point number

» `a @ b` (matrix multiplication)

» `a + b` (matrix addition)

» `c * a` (matrix scaling)

We've seen these, we've used them a bit, we'll use them much more

Analyzing Linear Algebra Algorithms

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We will not use $O(\cdot)$ notation!

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For numerics, we care about number of **F**loating-oint
Operations (FLOPs):

- » addition
- » subtraction
- » multiplication
- » division
- » square root

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$2n$ vs. n is very different
when $n \sim 10^{20}$

Dominant Terms

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That said, we don't care about *exact* bounds

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A function $f(n)$ is ***asymptotically equivalent*** to $g(n)$ if

$$\lim_{i \rightarrow \infty} \frac{f(i)}{g(i)} = 1$$

Dominant Terms

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A function $f(n)$ is ***asymptotically equivalent*** to $g(n)$ if

$$\lim_{i \rightarrow \infty} \frac{f(i)}{g(i)} = 1$$

for polynomials, they are equivalent to their dominant term

Dominant Terms

the dominant term of a polynomial is the monomial with the highest degree

$$\lim_{i \rightarrow \infty} \frac{3x^3 + 100000x^2}{3x^3} = 1$$

$3x^3$ dominates the function even though the coefficient for x^2 is so large

A Note on Complexity

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Suppose A and B are $n \times n$ matrices

This operations takes n multiplications and n divisions ($2n$ FLOPS total)

Repeating for each entry gives $\sim 2n^3$ FLOPS

A Note on Parallelization

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

The main part of this procedure is highly parallelizable

A Note on Parallelization

```
a = np.array(...)  
b = np.array(...)  
prod = np.zeros([a.shape[0], b.shape[1]])  
for i in range(a.shape[0]):  
    for j in range(b.shape[1]):  
        prod[i, j] = np.dot(a[i], b[:,j])
```

The main part of this procedure is highly parallelizable

One processor per entry gets you to $\sim 2n$ FLOPS

A Note on Libraries

There are lots of considerations when doing linear algebra with computers

Best leave it to experts (or do research in the area)

LAPACK is the state of the art library for matrix operations

numpy uses LAPACK

Summary

We can reason about matrix equations by reasoning directly about properties of linear transformations

Matrix multiplication coincides with composition of linear transformations

There is an algebra of matrices which is consistent with the algebra of vectors